DIVINE GRACE COLLEGE OF ARTS, SCIENCE AND TECHNOLOGY (DGCAST) – GAROUA

GEOGRAPHY REVISION SHEET – FORM FOUR

First Sequence Evaluation, School Year 2026-2027 • Teacher: Mr KAMDEM Emmanuel

Lessons revised: the Form 4 paper question by question: soils and erosion, ecosystems, hot deserts, Cameroon, climate table and competence

PART A: MULTIPLE CHOICE QUESTIONS 1 TO 10 (what each question asks)

1. MCQ 1: Soil, its composition and its profile.

Answer: Soil is the loose top layer of the Earth's surface in which plants grow. It is formed by physical, chemical and biological processes and is made of **mineral particles** (about 45 %), **organic matter** or humus (about 5 %), and **water and air** (about half). Profile: topsoil (A), subsoil (B), regolith (C), bedrock (R).

2. MCQ 2: Soil erosion and accelerated erosion; causes and consequences.

Answer: Soil erosion is the removal of the topsoil by water, wind or people. **Accelerated erosion** is erosion made very fast by human activities (deforestation, overgrazing, bush fires, ploughing down slopes); **normal (geologic) erosion** is slow and natural. Forms: sheet, rill, gully, wind. Consequences: loss of fertility, falling harvests, gullies, rivers and dams filled with mud.

3. MCQ 3: Soil conservation methods.

Answer: Terracing: flat steps cut into a steep slope. **Contour ploughing**: ploughing along the contours. **Strip cultivation**: strips of crops across the slope. **Windbreaks**: rows of trees against wind. Also: stone lines, cover crops, mulching, crop rotation, manure, controlled grazing, stopping bush fires.

4. MCQ 4, 5 and 6: The ecosystem, producers and energy flow.

Answer: Ecosystem: a natural area where living (biotic) and non-living (abiotic) components interact. Components: abiotic factors, **producers** (green plants that make their own food from solar energy: autotrophs), consumers (herbivores, carnivores), decomposers (fungi and bacteria). Energy: only about **10 %** passes from one trophic level to the next; the rest is lost as heat. Example of a food chain: grass → zebra → lion.

5. MCQ 7: Features of hot deserts. Which are depositional?

Answer: Hot deserts: less than 250 mm of rain, very hot days and cold nights. **Depositional** (wind deposits sand): **barchans** (crescent-shaped dunes) and seif (long) dunes. **Erosional** (wind wears rock): **inselbergs**, rock pedestals, yardangs, zeugen.

6. MCQ 8: Solving water scarcity in hot deserts.

Answer: Build **water reservoirs** and dams, dig wells and boreholes, bring water by pipes or canals from rivers, irrigate oasis farms, harvest rainwater, use drip irrigation. (Wearing long sleeves or drilling for oil does not give water.)

7. MCQ 9 and 10: Cameroon, relief and location.

Answer: Relief units: Western Highlands (Mount Cameroon, 4,095 m), Adamawa Plateau, Mandara Mountains, South Cameroon Plateau; lowlands: Coastal lowlands, Benue lowland, Chad Plain. Location: between about **2° N and 13° N** and **8° E and 16° E**, in Central Africa, on the Gulf of Guinea (about 475,000 km²).

Quick quiz	Answer
Soil components (4)	**Minerals, humus, water, air**
Energy passed to the next trophic level	**About 10 %**
Crescent-shaped desert dune	**Barchan**
Latitudes of Cameroon	**2° N to 13° N**

PART B: STRUCTURAL QUESTION (CLIMATE TABLE OF A STATION IN AFRICA)

8. How do you calculate the annual range of temperature, find the wettest month and name the biome?

Answer: Annual range = highest − lowest monthly temperature (show: 33 − 26 = 7 gives the method). Wettest month = month with the highest rainfall. Total rainfall = sum of the 12 months. **Biome**: look at the rainfall and the seasons, then give two reasons.

9. Which biome? Compare the three biomes of the course.

Answer: Equatorial: 2–3 °C range, rain in every month (1,500–2,500 mm). **Savanna (tropical wet and dry)**: hot all year (about 25 to 33 °C), one rainy season and a long dry season, 750–1,500 mm. **Hot desert**: below 250 mm, very large daily and annual ranges. Always give two reasons: temperature and rainfall (total and seasons).

Practice: study the climate data of a station W in Africa (approximate values).

Month	J	F	M	A	M	J	J	A	S	O	N	D
Temp. (°C)	25	27	29	29	27	25	24	24	24	25	26	25
Rain (mm)	0	2	15	60	140	170	200	230	180	70	5	0

10. Station W: calculate the annual range of temperature, name the wettest month and the biome with two reasons.

Answer: Range = 29 − 24 = **5 °C**. Wettest month: **August** (230 mm). Biome: **savanna (tropical wet and dry)**: hot all year (24 to 29 °C) with a small range, and one long rainy season (May to October) followed by a long dry season, with about 1,070 mm of rain.

11. Savanna: adaptation of the vegetation to the dry season.

Answer: The **baobab** stores water in its trunk; many trees shed their leaves; roots are deep and the bark is thick against fires; grasses dry up and grow again with the first rains. Location: about 5° to 15° N and S, including northern Cameroon; soils: red ferruginous.

PART C: SECTION B, COMPETENCE (A FARMER WHOSE FARM IS LOSING SOIL)

12. Situation: a poor widow farms a hill slope; after two years her yields drop and the soil is carried away by wind and heavy rain. What is the problem?

Answer: The farm is losing its soil through **soil erosion** (by wind and water), so the yields fall and the soil loses its fertility.

13. Task 2: state two causes of the problem.

Answer: Farming on a steep slope without protection (ploughing up and down); clearing the plant cover (deforestation, bush fires); heavy rain and strong winds on bare soil; over-cultivation without rest for the soil.

14. Task 3: as a geographer, suggest two measures.

Answer: Terracing or **contour ploughing** across the slope; planting trees, **windbreaks** and cover crops; mulching; crop rotation and manure to keep the soil fertile.

Exam tip: *Answer **every** question, write in complete words, show the working for the range of temperature, and keep two marks for **presentation** (clean, neat and correct spelling).*